Solving Ax = 0: Pivot Variables, Special Solutions

Lesson 6 said what the do-nothing trades are. This lesson finds every one of them, for any menu of funds: simplify A to U and then to R, and read the rank, the pivot and free variables, and every solution off R.

PART 1 Index

Note 1 of the pair: $Ax = 0$ and the null space. (Note 2 is Lesson 8, $Ax = b$.)

1. **Conditions and important concepts**
   13 short definitions: R, pivot, rank, pivot and free variables, special solutions, N, dimension.
2. **Procedure for Ax = 0**
   The 8 steps, in order.
3. **2 × 2 example**
   Nifty and its twin ETF: one free variable.
4. **3 × 3 example**
   Nifty, bonds, balanced: one free variable. And midcap: none.
5. **Bigger example: two free variables**
   Four funds, three rules: two special solutions, the matrix N, every solution. With a try-it box.

METHODS IN THIS LESSON: Row operations keep N(A) · Reduced row echelon form R · Rank · Pivot and free variables · n − rank · Special solutions · Nullspace matrix N · Echelon form U

PART 2 Prerequisites

Click any link to review it, then come back.

- The nullspace (Lesson 6): trades with $Ax = 0$, which change nothing.
- Elimination, 3 × 3 (Lesson 2) and Gauss–Jordan (Lesson 3).

PART 3

Explanation

THE FUNDS USED IN THE EXAMPLES

Fund	What ₹1 lakh does to (budget, return, exposure)
Nifty	$(1, 12, 1)$
Nifty ETF (a twin of Nifty)	$(1, 12, 1)$
Bonds	$(1, 7, 0)$
Balanced (half Nifty, half bonds)	$(1, 9.5, 0.5)$
Midcap	$(1, 15, 1.2)$

Each fund is a column of A; each rule is a row. A trade x lists ₹ lakh into each fund (positive = buy, negative = sell).

1. Conditions and important concepts

1. What is Ax = 0?

$Ax = 0$ means:

$$\boxed{\text{Find all trades } x \text{ that produce ZERO change in every rule.}}$$

Budget unchanged, return unchanged, exposure unchanged. The set of all such x is called the **null space**, $N(A)$.

2. What is R?

R = **Reduced Row Echelon Form** (RREF): the matrix we get after completely row-reducing A. We get there in two stages:

$$A \xrightarrow{\text{clear below the pivots}} U \xrightarrow{\text{pivots to 1, clear above}} R$$

METHOD · REDUCED ROW ECHELON FORM R

Eliminate below each pivot ($A \to U$), then divide each pivot row by its pivot and clear the entries above each pivot ($U \to R$). Every pivot becomes a 1, and it is the only non-zero entry in its column.

RULE · ROW OPERATIONS KEEP THE NULL SPACE

Row operations only rewrite the rules ("return minus 12 × budget"). A trade that gives zero for every old rule gives zero for every new one, and the steps can be undone. So $Ax = 0$ and $Rx = 0$ have **exactly the same solutions**. That's why we can solve with R.

IMPORTANT

R does **not** mean rank. R = Reduced Row Echelon Form. Rank = number of pivots.

3. What is a pivot?

A **pivot** is a leading 1 in R (the first non-zero entry of a row). Example: Nifty, bonds, balanced (worked out in the 3 × 3 example):

$$R = \begin{bmatrix} \boxed{1} & 0 & 0.5 \\ 0 & \boxed{1} & 0.5 \\ 0 & 0 & 0 \end{bmatrix} \qquad \begin{aligned} &\text{Pivot 1} \to \text{column 1 (Nifty)} \\ &\text{Pivot 2} \to \text{column 2 (bonds)} \end{aligned}$$

DEFINITION · RANK = NUMBER OF PIVOTS

Here there are 2 pivots, so $\text{rank} = 2$. In money terms: only 2 of these funds bring something new.

4. What is a pivot variable?

The variable belonging to a pivot column. Above: x_1 (Nifty) and x_2 (bonds) are pivot variables.

5. What is a free variable?

A variable whose column does **not** contain a pivot. Above: x_3 (balanced) is a free variable.

RULE · PIVOT VARIABLES AND FREE VARIABLES

Pivot column → pivot variable (decided by its row of R). Column without a pivot → free variable (you choose it).

6. What does a free variable mean?

A free variable is a variable whose value **we can choose**. The rows of R above say $x_1 + 0.5x_3 = 0$ and $x_2 + 0.5x_3 = 0$, so once we choose x_3, the pivot variables are decided:

Choose x_3 (balanced)	Then $x_1 = -0.5x_3$ (Nifty)	And $x_2 = -0.5x_3$ (bonds)	Trade
0	0	0	$(0, 0, 0)$: do nothing
1	-0.5	-0.5	$(-0.5, -0.5, 1)$
5	-2.5	-2.5	$(-2.5, -2.5, 5)$
-3	1.5	1.5	$(1.5, 1.5, -3)$

All are possible choices. In money terms: buy some balanced, sell half that amount in each of Nifty and bonds. Nothing changes.

7. Number of free variables

> **RULE · FREE VARIABLES = N − RANK**
>
> With n variables (funds): $\text{number of free variables} = n - \text{rank}$. Above: $3 - 2 = 1$.

8. Special solutions

> **METHOD · SPECIAL SOLUTIONS**
>
> Give **one** free variable the value 1 and all other free variables the value 0, then solve for the pivot variables. Each free variable gives **one** special solution.

$$\boxed{\text{Number of special solutions} = \text{number of free variables}}$$

Above: $x_3 = 1$ gives $s_1 = (-0.5, -0.5, 1)$.

9. When do special solutions exist?

If the number of free variables is **more than 0**: non-zero special solutions exist (Nifty, bonds, balanced: 1 free variable).

If the number of free variables is **0**: there are no non-zero special solutions, and the only solution is

$$\boxed{x = 0}$$

Example: Nifty, bonds, midcap reduces to $R = I$ (3 pivots, 0 free variables), so the only do-nothing trade is doing nothing (worked in the 3 × 3 example).

10. Nullspace matrix N

> **METHOD · NULLSPACE MATRIX N**
>
> Put all special solutions side by side as **columns**: if they are $s_1, s_2, \ldots, s_k$, then $N = \begin{bmatrix} s_1 & s_2 & \cdots & s_k \end{bmatrix}$.

Above there is one special solution, so N has one column: $N = \begin{bmatrix} -0.5 \\ -0.5 \\ 1 \end{bmatrix}$.

11. Important null space property

Every column of N is a null space solution. Therefore:

$$\boxed{AN = 0}$$

Check above, row by row of $A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix}$: budget $-0.5 - 0.5 + 1 = 0$ ✓; return $-6 - 3.5 + 9.5 = 0$ ✓; exposure $-0.5 + 0 + 0.5 = 0$ ✓.

12. All null space solutions

If there are two special solutions s_1 and s_2, every null space solution is

$$\boxed{x = c\,s_1 + d\,s_2}$$

where c and d can be **any** numbers. (With one special solution: $x = c\,s_1$.)

13. Null space dimension

$$\boxed{\text{Dimension of null space} = \text{number of free variables} = n - \text{rank}}$$

Dimension 1: a line of trades. Dimension 2: a plane. Dimension 0: just the point $x = 0$.

2. Procedure for Ax = 0

Step	What to do
1	Row reduce $A \to R$ (first $A \to U$, then $U \to R$).
2	Find the pivot columns.
3	Identify the free variables (columns without a pivot).
4	For each free variable: make it $= 1$ and all other free variables $= 0$.
5	Solve for the pivot variables using the rows of R.
6	Each result is a special solution.
7	Put the special solutions side by side: $N = \begin{bmatrix} s_1 & s_2 & \cdots \end{bmatrix}$.
8	All null space solutions are combinations of those special solutions.

3. 2 × 2 example

Nifty and its twin ETF, with the budget and return rules. We want $Ax = 0$:

$$A = \begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix} \begin{matrix} \leftarrow \text{budget} \\ \leftarrow \text{return} \end{matrix}$$

Step 1, row reduce. Row 2 $-$ 12$\times$ row 1: $12 - 12(1) = 0,\ 12 - 12(1) = 0$. The pivot is already 1, nothing above it:

$$R = \begin{bmatrix} \boxed{1} & 1 \\ 0 & 0 \end{bmatrix}$$

Steps 2–3. Pivot: column 1. Free variable: x_2 (ETF).

Equation (row 1 of R):

$$x_1 + x_2 = 0$$

Steps 4–5. Set the free variable $x_2 = 1$. Then $x_1 + 1 = 0$, so $x_1 = -1$.

Step 6, special solution:

$$\boxed{s_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}} \qquad \text{sell ₹1 lakh Nifty, buy ₹1 lakh ETF}$$

Steps 7–8. Nullspace matrix and every solution:

$$N = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \qquad x = c \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Check $AN = 0$: budget $1(-1) + 1(1) = 0$ ✓; return $12(-1) + 12(1) = 0$ ✓.

4. 3 × 3 example

Nifty, bonds, balanced, with all three rules:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix} \begin{matrix} \leftarrow \text{budget} \\ \leftarrow \text{return} \\ \leftarrow \text{exposure} \end{matrix}$$

Step 1a, A → U (clear below the pivots):

- Row 2 $-$ 12$\times$ row 1: $12 - 12 = 0,\ 7 - 12 = -5,\ 9.5 - 12 = -2.5 \to (0, -5, -2.5)$.
- Row 3 $-$ 1$\times$ row 1: $1 - 1 = 0,\ 0 - 1 = -1,\ 0.5 - 1 = -0.5 \to (0, -1, -0.5)$.
- Row 3 $-$ 0.2$\times$ row 2 (multiplier $(-1)/(-5) = 0.2$): $0,\ -1 + 1 = 0,\ -0.5 + 0.5 = 0 \to (0, 0, 0)$.

> **METHOD · ECHELON FORM U (THE STAIRCASE)**
>
> After clearing below every pivot, each row starts further right than the row above, and all-zero rows sit at the bottom.

$$U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & -2.5 \\ 0 & 0 & 0 \end{bmatrix}$$

Step 1b, U → R (pivots to 1, clear above):

- Row 2 $\div(-5)$: $0,\ 1,\ -2.5/(-5) = 0.5 \to (0, 1, 0.5)$.
- Row 1 $-$ 1$\times$ row 2: $1 - 0 = 1,\ 1 - 1 = 0,\ 1 - 0.5 = 0.5 \to (1, 0, 0.5)$.

$$R = \begin{bmatrix} \boxed{1} & 0 & 0.5 \\ 0 & \boxed{1} & 0.5 \\ 0 & 0 & 0 \end{bmatrix}$$

Steps 2–3. Pivot columns: 1 and 2. Free column: 3. So x_1 = pivot (Nifty), x_2 = pivot (bonds), x_3 = free (balanced). Rank $= 2$; free variables $= 3 - 2 = 1$.

Equations (rows of R):

$$x_1 + 0.5x_3 = 0$$
$$x_2 + 0.5x_3 = 0$$

Steps 4–5. Set $x_3 = 1$. Then $x_1 = -0.5$ and $x_2 = -0.5$.

Step 6, special solution:

$$\boxed{s_1 = \begin{bmatrix} -0.5 \\ -0.5 \\ 1 \end{bmatrix}} \quad \text{buy ₹1 lakh balanced, sell ₹0.5 lakh each of Nifty and bonds}$$

Steps 7–8. $N = s_1$ and every solution is $x = c\,s_1$ (Lesson 6's $(1, 1, -2)$ is $c = -2$).

Check $AN = 0$: budget $-0.5 - 0.5 + 1 = 0$ ✓; return $12(-0.5) + 7(-0.5) + 9.5(1) = -6 - 3.5 + 9.5 = 0$ ✓; exposure $-0.5 + 0 + 0.5 = 0$ ✓.

Same size, no free variable: Nifty, bonds, midcap

- A → U: row 2 $-\ 12\times$ row 1 $\to (0, -5, 3)$; row 3 $-$ row 1 $\to (0, -1, 0.2)$; row 3 $-\ 0.2\times$ row 2 $\to (0, 0, 0.2 - 0.6) = (0, 0, -0.4)$.
- U → R: row 3 $\div(-0.4) \to (0, 0, 1)$; row 2 $-\ 3\times$ row 3 $\to (0, -5, 0)$, then $\div(-5) \to (0, 1, 0)$; row 1 $-$ row 2 $-$ row 3 $\to (1, 0, 0)$.

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad 3 \text{ pivots, } 0 \text{ free variables } \Rightarrow \boxed{x = 0 \text{ only}}$$

Midcap brings something new, so no fund can be rebuilt from the others.

5. Bigger example: two free variables

Four funds (Nifty, ETF, bonds, balanced), three rules:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 \\ 1 & 1 & 0 & 0.5 \end{bmatrix}$$

Step 1a, A → U:

- Row 2 $-\,12\times$ row 1: $0,\ 0,\ 7-12=-5,\ 9.5-12=-2.5 \to (0,0,-5,-2.5)$.
- Row 3 $-\,1\times$ row 1: $0,\ 0,\ 0-1=-1,\ 0.5-1=-0.5 \to (0,0,-1,-0.5)$.
- Column 2 has nothing left to use as a pivot (the ETF column was cleared along with Nifty), so skip it.
- Row 3 $-\,0.2\times$ row 2: $0,\ 0,\ -1+1=0,\ -0.5+0.5=0 \to (0,0,0,0)$.

Step 1b, U → R:

- Row 2 $\div(-5)$: $(0,0,1,0.5)$.
- Row 1 $-\,1\times$ row 2: $1,\ 1,\ 1-1=0,\ 1-0.5=0.5 \to (1,1,0,0.5)$.

$$R=\begin{bmatrix}\boxed{1} & 1 & 0 & 0.5\\ 0 & 0 & \boxed{1} & 0.5\\ 0 & 0 & 0 & 0\end{bmatrix}$$

Steps 2–3. Pivot variables: x_1 (Nifty), x_3 (bonds). Free variables: x_2 (ETF), x_4 (balanced). Rank $=2$; free variables $=4-2=2$.

Equations:

$$\begin{aligned} x_1+x_2+0.5x_4&=0\\ x_3+0.5x_4&=0\end{aligned}$$

Special solution 1

$x_2=1, x_4=0$. Then $x_1+1+0=0 \Rightarrow x_1=-1$, and $x_3+0=0 \Rightarrow x_3=0$.

$$\boxed{s_1=\begin{bmatrix}-1\\1\\0\\0\end{bmatrix}}$$ swap ₹1 lakh Nifty for the ETF

Special solution 2

$x_2=0, x_4=1$. Then $x_1+0+0.5=0 \Rightarrow x_1=-0.5$, and $x_3+0.5=0 \Rightarrow x_3=-0.5$.

$$\boxed{s_2=\begin{bmatrix}-0.5\\0\\-0.5\\1\end{bmatrix}}$$ build ₹1 lakh balanced from Nifty and bonds

Nullspace matrix

$$N = \begin{bmatrix} -1 & -0.5 \\ 1 & 0 \\ 0 & -0.5 \\ 0 & 1 \end{bmatrix} \begin{matrix} \leftarrow \text{Nifty} \\ \leftarrow \text{ETF} \\ \leftarrow \text{bonds} \\ \leftarrow \text{balanced} \end{matrix}$$

Check $AN = 0$	$s_1 = (-1, 1, 0, 0)$	$s_2 = (-0.5, 0, -0.5, 1)$
Budget $(1, 1, 1, 1)$	$-1 + 1 + 0 + 0 = 0$ ✓	$-0.5 + 0 - 0.5 + 1 = 0$ ✓
Return $(12, 12, 7, 9.5)$	$-12 + 12 + 0 + 0 = 0$ ✓	$-6 + 0 - 3.5 + 9.5 = 0$ ✓
Exposure $(1, 1, 0, 0.5)$	$-1 + 1 + 0 + 0 = 0$ ✓	$-0.5 + 0 + 0 + 0.5 = 0$ ✓

Every null space solution

$$\boxed{x = c\,s_1 + d\,s_2} \qquad \text{dimension 2: a plane of do-nothing trades}$$

Example: $c = 2, d = 2$ gives $(-2 - 1,\ 2,\ -1,\ 2) = (-3, 2, -1, 2)$. Budget $-3 + 2 - 1 + 2 = 0$ ✓, return $-36 + 24 - 7 + 19 = 0$ ✓, exposure $-3 + 2 + 0 + 1 = 0$ ✓.

Try it: choose c and d

ETF amount $x_2 = c$ 2 Balanced amount $x_4 = d$ 2

Free (you chose): ETF $x_2 = 2$, balanced $x_4 = 2$.
Forced: Nifty $x_1 = -c - 0.5d = -3$, bonds $x_3 = -0.5d = -1$.

Trade $= 2\,s_1 + 2\,s_2 = (-3, 2, -1, 2)$ (₹ lakh in Nifty, ETF, bonds, balanced).

Budget: $1(-3) + 1(2) + 1(-1) + 1(2) = 0$ ✓
Return: $12(-3) + 12(2) + 7(-1) + 9.5(2) = 0$ ✓
Exposure: $1(-3) + 1(2) + 0(-1) + 0.5(2) = 0$ ✓

Nothing changes: every rule stays at 0, whatever free amounts you pick.

▼ Check yourself (MIT problem 7.1): find R, the rank and the special solutions of $\begin{bmatrix} 1 & 5 & 7 & 9 \\ 0 & 4 & 1 & 7 \\ 2 & -2 & 11 & -3 \end{bmatrix}$.

A → U: row 3 $-$ 2× row 1 $= (0, -12, -3, -21)$; row 3 $+$ 3× row 2 $= (0, 0, 0, 0)$. U → R: row 2 ÷4 $= (0, 1, \frac{1}{4}, \frac{7}{4})$; row 1 $-$ 5× row 2 $= (1, 0, \frac{23}{4}, \frac{1}{4})$. Rank 2; free x_3, x_4. $s_1 = (-\frac{23}{4}, -\frac{1}{4}, 1, 0)$, $s_2 = (-\frac{1}{4}, -\frac{7}{4}, 0, 1)$. Try it in the lab on the Examples tab.